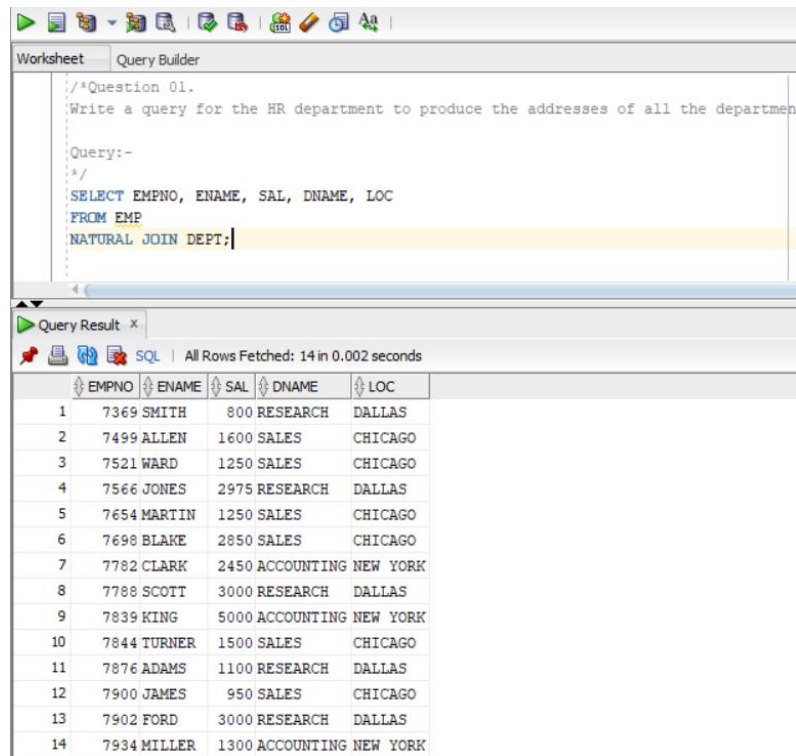


Displaying Data from Multiple Tables using Joins

Handson assessment Questions----

01.



Worksheet Query Builder

/*Question 01.
Write a query for the HR department to produce the addresses of all the department

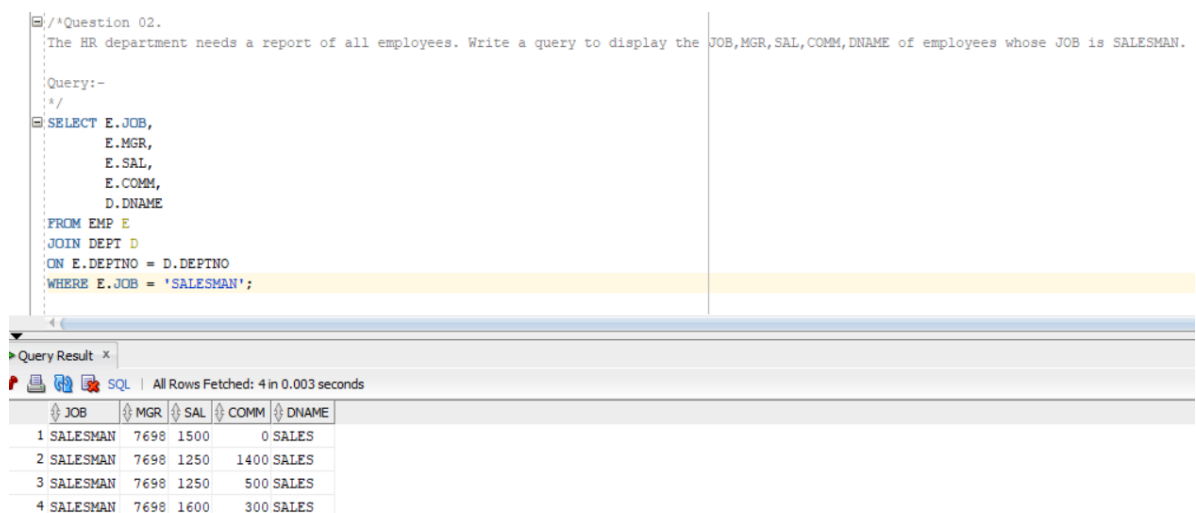
Query:-
*/
SELECT EMPNO, ENAME, SAL, DNAME, LOC
FROM EMP
NATURAL JOIN DEPT;

Query Result x

All Rows Fetched: 14 in 0.002 seconds

	EMPNO	ENAME	SAL	DNAME	LOC
1	7369	SMITH	800	RESEARCH	DALLAS
2	7499	ALLEN	1600	SALES	CHICAGO
3	7521	WARD	1250	SALES	CHICAGO
4	7566	JONES	2975	RESEARCH	DALLAS
5	7654	MARTIN	1250	SALES	CHICAGO
6	7698	BLAKE	2850	SALES	CHICAGO
7	7782	CLARK	2450	ACCOUNTING	NEW YORK
8	7788	SCOTT	3000	RESEARCH	DALLAS
9	7839	KING	5000	ACCOUNTING	NEW YORK
10	7844	TURNER	1500	SALES	CHICAGO
11	7876	ADAMS	1100	RESEARCH	DALLAS
12	7900	JAMES	950	SALES	CHICAGO
13	7902	FORD	3000	RESEARCH	DALLAS
14	7934	MILLER	1300	ACCOUNTING	NEW YORK

02.



/*Question 02.
The HR department needs a report of all employees. Write a query to display the JOB,MGR,SAL,COMM,DNAME of employees whose JOB is SALESMAN.

Query:-
*/
SELECT E.JOB,
E.MGR,
E.SAL,
E.COMM,
D.DNAME
FROM EMP E
JOIN DEPT D
ON E.DEPTNO = D.DEPTNO
WHERE E.JOB = 'SALESMAN';

Query Result x

All Rows Fetched: 4 in 0.003 seconds

	JOB	MGR	SAL	COMM	DNAME
1	SALESMAN	7698	1500	0	SALES
2	SALESMAN	7698	1250	1400	SALES
3	SALESMAN	7698	1250	500	SALES
4	SALESMAN	7698	1600	300	SALES

03.

```
/*Question 03.
The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.

Query:-
*/
SELECT E.ENAME,
       E.JOB,
       E.DEPTNO,
       D.DNAME
FROM EMP E
JOIN DEPT D
ON E.DEPTNO = D.DEPTNO
WHERE D.LOC = 'DALLAS';
```

Query Result x

SQL | All Rows Fetched: 5 in 0.004 seconds

	ENAME	JOB	DEPTNO	DNAME
1	SMITH	CLERK	20	RESEARCH
2	JONES	MANAGER	20	RESEARCH
3	SCOTT	ANALYST	20	RESEARCH
4	ADAMS	CLERK	20	RESEARCH
5	FORD	ANALYST	20	RESEARCH

04.

```
/*Question 04.
Create a report to display employees' ename and employee number along with their manager'

Query:-
*/
SELECT E.ENAME AS "Employee",
       E.EMPNO AS "Emp#",
       M.ENAME AS "Manager",
       M.EMPNO AS "Mgr#"
FROM EMP E
JOIN EMP M
ON E.MGR = M.EMPNO;
```

Query Result x

SQL | All Rows Fetched: 13 in 0.004 seconds

	Employee	Emp#	Manager	Mgr#
1	FORD	7902	JONES	7566
2	SCOTT	7788	JONES	7566
3	JAMES	7900	BLAKE	7698
4	TURNER	7844	BLAKE	7698
5	MARTIN	7654	BLAKE	7698
6	WARD	7521	BLAKE	7698
7	ALLEN	7499	BLAKE	7698
8	MILLER	7934	CLARK	7782
9	ADAMS	7876	SCOTT	7788
10	CLARK	7782	KING	7839
11	BLAKE	7698	KING	7839
12	JONES	7566	KING	7839
13	SMITH	7369	FORD	7902

05.

```

/*Question 05.
Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.

Query:-
*/
SELECT E.ENAME AS "Employee",
       E.EMPNO AS "Emp#",
       M.ENAME AS "Manager",
       M.EMPNO AS "Mgr#"
FROM EMP E
LEFT OUTER JOIN EMP M
ON E.MGR = M.EMPNO
ORDER BY E.EMPNO;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.003 seconds

	Employee	Emp#	Manager	Mgr#
1	SMITH	7369	FORD	7902
2	ALLEN	7499	BLAKE	7698
3	WARD	7521	BLAKE	7698
4	JONES	7566	KING	7839
5	MARTIN	7654	BLAKE	7698
6	BLAKE	7698	KING	7839
7	CLARK	7782	KING	7839
8	SCOTT	7788	JONES	7566
9	KING	7839	(null)	(null)
10	TURNER	7844	BLAKE	7698
11	ADAMS	7876	SCOTT	7788
12	JAMES	7900	BLAKE	7698
13	FORD	7902	JONES	7566
14	MILLER	7934	CLARK	7782

06.

```

/*Question 06.
The HR department needs a report on job grades and sala

Query:-
*/
SELECT E.ENAME,
       E.JOB,
       D.DNAME,
       E.SAL,
       S.GRADE
FROM EMP E
JOIN DEPT D
ON E.DEPTNO = D.DEPTNO
JOIN SALGRADE S
ON E.SAL BETWEEN S.LOSAL AND S.HISAL;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.028 seconds

	ENAME	JOB	DNAME	SAL	GRADE
1	SMITH	CLERK	RESEARCH	800	1
2	ALLEN	SALESMAN	SALES	1600	3
3	WARD	SALESMAN	SALES	1250	2
4	JONES	MANAGER	RESEARCH	2975	4
5	MARTIN	SALESMAN	SALES	1250	2
6	BLAKE	MANAGER	SALES	2850	4
7	CLARK	MANAGER	ACCOUNTING	2450	4
8	SCOTT	ANALYST	RESEARCH	3000	4
9	KING	PRESIDENT	ACCOUNTING	5000	5
10	TURNER	SALESMAN	SALES	1500	3
11	ADAMS	CLERK	RESEARCH	1100	1
12	JAMES	CLERK	SALES	950	1
13	FORD	ANALYST	RESEARCH	3000	4

07.

```

/*Question 07.
Display the ENAME,DNAME of all the employees. Also display those department name which do not have any employees working.

Query:-
*/
SELECT E.ENAME,
       D.DNAME
FROM EMP E
RIGHT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;

```

Query Result x

SQL | All Rows Fetched: 15 in 0.002 seconds

	ENAME	DNAME
1	SMITH	RESEARCH
2	ALLEN	SALES
3	WARD	SALES
4	JONES	RESEARCH
5	MARTIN	SALES
6	BLAKE	SALES
7	CLARK	ACCOUNTING
8	SCOTT	RESEARCH
9	KING	ACCOUNTING
10	TURNER	SALES
11	ADAMS	RESEARCH
12	JAMES	SALES
13	FORD	RESEARCH
14	MILLER	ACCOUNTING
15	(null)	OPERATIONS

08.

```

/*Question 08.
The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

Query:-
*/
SELECT E.ENAME,
       E.HIREDATE,
       M.ENAME,
       M.HIREDATE
FROM EMP E
JOIN EMP M
ON E.MGR = M.EMPNO
WHERE E.HIREDATE < M.HIREDATE;

```

Query Result x

SQL | All Rows Fetched: 6 in 0.005 seconds

	ENAME	HIREDATE	ENAME_1	HIREDATE_1
1	WARD	22-02-81	BLAKE	01-05-81
2	ALLEN	20-02-81	BLAKE	01-05-81
3	CLARK	09-06-81	KING	17-11-81
4	BLAKE	01-05-81	KING	17-11-81
5	JONES	02-04-81	KING	17-11-81
6	SMITH	17-12-80	FORD	03-12-81

09.

```
/*Question 09.
Display the EMPNO,ENAME,DNAME,LOC of those employees who are working as CLERK. Use the USING clause.

Query:-
*/
SELECT EMPNO,
       ENAME,
       DNAME,
       LOC
FROM EMP
JOIN DEPT
USING (DEPTNO)
WHERE JOB = 'CLERK';
```

Query Result x

SQL | All Rows Fetched: 4 in 0.004 seconds

	EMPNO	ENAME	DNAME	LOC
1	7934	MILLER	ACCOUNTING	NEW YORK
2	7876	ADAMS	RESEARCH	DALLAS
3	7369	SMITH	RESEARCH	DALLAS
4	7900	JAMES	SALES	CHICAGO

10.

```
/*Question 10.
Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000. Use the ON clause.

Query:-
*/
SELECT E.ENAME,
       E.SAL,
       E.MGR,
       D.DNAME
FROM EMP E
JOIN DEPT D
ON E.DEPTNO = D.DEPTNO
WHERE E.SAL > 2000;
```

Query Result x

SQL | All Rows Fetched: 6 in 0.004 seconds

	ENAME	SAL	MGR	DNAME
1	JONES	2975	7839	RESEARCH
2	BLAKE	2850	7839	SALES
3	CLARK	2450	7839	ACCOUNTING
4	SCOTT	3000	7566	RESEARCH
5	KING	5000	(null)	ACCOUNTING
6	FORD	3000	7566	RESEARCH

11.

```

/*Question 11.
Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees. Use LEFT OUTER JOIN.

Query:-
*/
SELECT E.EMPNO,
       E.ENAME,
       E.JOB,
       D.DEPTNO,
       D.DNAME,
       D.LOC
FROM EMP E
LEFT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.003 seconds

	EMPNO	ENAME	JOB	DEPTNO	DNAME	LOC
1	7934	MILLER	CLERK	10	ACCOUNTING	NEW YORK
2	7839	KING	PRESIDENT	10	ACCOUNTING	NEW YORK
3	7782	CLARK	MANAGER	10	ACCOUNTING	NEW YORK
4	7902	FORD	ANALYST	20	RESEARCH	DALLAS
5	7876	ADAMS	CLERK	20	RESEARCH	DALLAS
6	7788	SCOTT	ANALYST	20	RESEARCH	DALLAS
7	7566	JONES	MANAGER	20	RESEARCH	DALLAS
8	7369	SMITH	CLERK	20	RESEARCH	DALLAS
9	7900	JAMES	CLERK	30	SALES	CHICAGO
10	7844	TURNER	SALESMAN	30	SALES	CHICAGO
11	7698	BLAKE	MANAGER	30	SALES	CHICAGO
12	7654	MARTIN	SALESMAN	30	SALES	CHICAGO
13	7521	WARD	SALESMAN	30	SALES	CHICAGO
14	7499	ALLEN	SALESMAN	30	SALES	CHICAGO

12.

```

/*Question 12.
Display the ENAME,DNAME of employees. Use RIGHT OUTER JOIN.

Query:-
*/
SELECT E.ENAME,
       D.DNAME
FROM EMP E
RIGHT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;

```

Query Result x

SQL | All Rows Fetched: 15 in 0.002 seconds

	ENAME	DNAME
1	SMITH	RESEARCH
2	ALLEN	SALES
3	WARD	SALES
4	JONES	RESEARCH
5	MARTIN	SALES
6	BLAKE	SALES
7	CLARK	ACCOUNTING
8	SCOTT	RESEARCH
9	KING	ACCOUNTING
10	TURNER	SALES
11	ADAMS	RESEARCH
12	JAMES	SALES
13	FORD	RESEARCH
14	MILLER	ACCOUNTING
15	(null)	OPERATIONS

13.

```
/*Question 13.
Display the EMPNO,DNAME,LOC of employees. Use FULL OUTER JOIN.
Query:-
*/
SELECT E.EMPNO,
       D.DNAME,
       D.LOC
FROM EMP E
FULL OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;
```

Query Result x

SQL | All Rows Fetched: 15 in 0.003 seconds

	EMPNO	DNAME	LOC
1	7369	RESEARCH	DALLAS
2	7499	SALES	CHICAGO
3	7521	SALES	CHICAGO
4	7566	RESEARCH	DALLAS
5	7654	SALES	CHICAGO
6	7698	SALES	CHICAGO
7	7782	ACCOUNTING	NEW YORK
8	7788	RESEARCH	DALLAS
9	7839	ACCOUNTING	NEW YORK
10	7844	SALES	CHICAGO
11	7876	RESEARCH	DALLAS
12	7900	SALES	CHICAGO
13	7902	RESEARCH	DALLAS
14	7934	ACCOUNTING	NEW YORK
15	(null)	OPERATIONS	BOSTON